Notes: Bianchi, Ludvigson, and Ma (AER, 2022)

Belief Distortions and Macroeconomic Fluctuations

Contents

1 Big picture

- **Question.** How large are systematic expectational errors in macroeconomic survey forecasts, and how do those belief distortions move with macroeconomic fluctuations?
- **Core idea.** The paper defines belief distortions as ex ante systematic forecast errors generated by misweighting available information that is relevant for prediction accuracy.
- **Main empirical challenge.** There is no obvious objective benchmark for nondistorted beliefs. Survey respondents operate in a data-rich real-time environment, and a benchmark that uses too little information, ex post data, or in-sample fitting can mismeasure distortions.
- **Method.** The authors build a dynamic machine learning benchmark that uses only real-time information available before survey deadlines. The machine observes the survey forecast itself, hundreds of public macro-financial variables summarized through factors, and then chooses shrinkage, sparsity, and rolling windows out of sample.
- **Surveys studied.** The paper examines the Survey of Professional Forecasters (SPF), the Michigan Survey of Consumers (SOC), and Blue Chip (BC) forecasts.
- **Variables studied.** The focus is four-quarter-ahead inflation and real GDP growth.
- **Main result 1.** The machine benchmark beats survey forecasts across surveys, variables, and respondent types, sometimes by large margins.
- **Main result 2.** All respondent types place too much weight on the private or judgmental component of their own forecasts relative to what can be learned from objective public information.
- **Main result 3.** Forecasts of both inflation and GDP growth oscillate between optimism and pessimism over time; distortions are especially visible after the Great Recession and in the final years of the evaluation sample.
- **Economic takeaway.** The paper treats machine learning as a real-time diagnostic tool for human judgment. It argues that artificial intelligence can identify and correct systematic human forecast distortions without relying on hindsight.

2 Incremental contribution of the paper

- **A new benchmark for belief distortions.** Earlier empirical work often evaluates survey forecasts against rational-expectations benchmarks built from small information sets, in-sample regressions, or ex post outcome data. This paper builds a real-time, out-of-sample, data-rich benchmark.
- **Machine learning as an expectation benchmark.** The paper uses dynamic machine learning not merely as a forecasting tool, but as a benchmark for what a nondistorted forecast could have achieved using information available at the time.
- **Controls for forecaster judgment.** The machine conditions on the current survey forecast. This matters because professional forecasts may include private information or judgment that is not directly observable to the econometrician.

- **Dynamic model selection.** Instead of selecting one fixed forecasting model with hindsight, the algorithm reselects the estimation window, validation window, shrinkage, and sparsity at each date.
- **Cross-survey and cross-type evidence.** The paper applies the same methodology across professional forecasters, households, and Blue Chip forecasters, and across different percentiles of each forecast distribution.
- **Quantifies the source of errors.** The bias decomposition shows that much of the error comes from overreliance on the survey respondent's own forecast and underweighting of public information.
- **Reinterprets earlier behavioral-macro evidence.** The paper shows that some evidence for underreaction and overreaction is much smaller when the benchmark is formed in real time and out of sample.
- **Business-cycle contribution.** The paper links belief distortions to cyclical shocks and shows that survey beliefs initially underreact and later overreact, but less dramatically than suggested by benchmarks that use ex post realized outcomes.
- **Practical contribution.** The analysis suggests that institutions could use machine algorithms to audit, discipline, and improve professional forecasts.

3 Data and measurement

3.1 Survey expectations

- **SPF.** The Survey of Professional Forecasters provides quarterly forecasts from professional forecasters. The paper uses forecasts for the GDP deflator and real GDP, and converts these into inflation and real GDP growth expectations.
- **SOC.** The Michigan Survey of Consumers measures household inflation expectations directly. For GDP growth, the paper converts qualitative household expectations about business conditions into a quantitative forecast.
- **BC.** The Blue Chip survey provides professional forecasts of real GDP growth and the GDP deflator at monthly frequency. The paper aligns the survey timing with quarterly horizons.
- The main forecasts are four-quarter-ahead expectations of inflation and real GDP growth.
- The authors analyze not only the mean or median forecast, but also respondent types defined by percentiles of the forecast distribution.

Interpretation: The percentile-type approach lets the authors study heterogeneous beliefs over long samples even though individual survey panelists enter and exit the surveys.

3.2 Information sets

- **Real-time macro data.** The paper constructs real-time macro variables from data vintages available before the survey deadline. The real-time macro panel includes 92 variables.

- **Monthly financial data.** The authors use 147 monthly financial indicators, including valuation ratios, default spreads, term spreads, bond yields, Treasury yields, and industry equity returns.
- **Daily financial data.** The paper adds daily information observed up to one day before the survey deadline, including commodities, credit spreads, equities, exchange rates, and government securities.
- **Nonfactor information.** The machine also sees nowcasts, lagged forecasts, lagged percentiles of forecast distributions, cross-sectional moments of forecasts, trend inflation, and detrended macro variables.
- After extracting factors and adding nonfactor information, the machine entertains 68 predictors for inflation and 72 for GDP growth before sparsity is imposed.

Interpretation: The central measurement contribution is the careful construction of what an informed respondent could have known in real time, rather than what the econometrician knows later.

3.3 Belief distortion measure

Let $\mathbb{F}_t^{(i)}[y_{j,t+h}]$ denote the survey forecast of outcome $y_{j,t+h}$ made at time t by respondent type i. Let $\mathbb{E}_t^{(i)}[y_{j,t+h}]$ denote the corresponding machine forecast. The paper defines the ex ante belief distortion as

$$\text{bias}_{j,t}^{(i)} = \mathbb{F}_t^{(i)}[y_{j,t+h}] - \mathbb{E}_t^{(i)}[y_{j,t+h}].$$

- A positive GDP-growth bias means optimism: the survey forecast is above the machine forecast.
- A positive inflation bias is labeled pessimism because respondents expect inflation to be higher than the machine benchmark.
- The object is ex ante bias, not ex post forecast error. The machine may also be wrong ex post.

Interpretation: The paper separates bias from bad luck. A large forecast mistake during an unforeseen event is not automatically a belief distortion.

4 Models and empirical specifications

4.1 Machine-efficient benchmark

The core forecasting specification is

$$y_{j,t+h} = \alpha_{jh}^{(i)} + \beta_{jh,\mathbb{F}}^{(i)}\mathbb{F}_t^{(i)}[y_{j,t+h}] + B_{jh,\mathcal{Z}}^{(i)\prime}\mathcal{Z}_{jt} + \epsilon_{jt+h},$$

where $\mathcal{Z}_{jt}$ collects real-time public information and estimated factors.

The machine-efficient benchmark corresponds to

$$\beta_{jh,\mathbb{F}}^{(i)} = 1, \qquad B_{jh,\mathcal{Z}}^{(i)} = 0, \qquad \alpha_{jh}^{(i)} = 0.$$

- If these restrictions hold, the survey forecast already processes the available information as efficiently as the machine.

- If $\beta^{(i)}_{jh,\mathbb{F}} < 1$, the survey respondent places too much weight on the marginal information in the survey forecast relative to public information.
- If $B^{(i)}_{jh,\mathcal{Z}} \neq 0$, the survey respondent underweights some public information conditional on the survey forecast.

Interpretation: This specification treats the survey forecast as containing both public information and private judgment. The machine asks whether that composite forecast is weighted efficiently.

4.2 Private and public signal interpretation

The paper gives a simple signal model. A forecaster combines a public statistical signal with a private or judgmental signal:

$$\mathbb{F} = \gamma_F z + (1 - \gamma_F)\alpha_F x.$$

The machine observes $\mathbb{F}$ and x, but not z, and estimates

$$\mathbb{E} = \widehat{\beta}\mathbb{F} + \widehat{B}x.$$

- If $\widehat{\beta} < 1$, the forecaster overrelies on private information or judgment relative to public information.
- This can happen because the forecaster overestimates the precision of the private signal or inefficiently combines the public signal.
- The private component can still be valuable; the issue is that respondents weight it too heavily.

Interpretation: The key behavioral message is not that forecasters have useless judgment. Rather, they appear to put too much relative weight on judgment.

4.3 Dynamic elastic-net learning algorithm

- The machine uses an elastic-net estimator, combining ridge shrinkage and lasso sparsity.
- At each forecast date, the prior sample is split into an estimation window and a validation window.
- The machine chooses the regularization parameters and the window lengths using pseudo out-of-sample validation.
- It then makes a genuine out-of-sample forecast for the external evaluation sample.
- The procedure is repeated recursively over time.

Interpretation: The machine is not allowed to know the future, choose variables with hindsight, or use final revised data. This makes the benchmark conservative.

4.4 Forecast comparison metrics

The key metric is the ratio of machine mean squared forecast error to survey mean squared forecast error:

$$\frac{MSE_{\mathbb{E}}}{MSE_{\mathbb{F}}}.$$

- Ratios below one indicate that the machine forecast is more accurate.
- The authors also report an out-of-sample R^2:

$$R^2_{OOS} = 1 - \frac{MSE_{\mathbb{E}}}{MSE_{\mathbb{F}}}.$$

- They compute optimal hybrid forecast weights to show whether the machine's improvement is economically meaningful.

Interpretation: The paper focuses on economic forecast improvement rather than only on binary statistical rejection tests.

5 Identification and empirical design

5.1 What the design identifies

- The paper identifies systematic ex ante forecast distortions relative to a machine benchmark using only information available before survey deadlines.
- The exercise does not prove that the machine forecast equals the true rational-expectations forecast.
- The evidence is strongest as a diagnostic statement: if the machine repeatedly improves prediction out of sample, survey respondents must have systematically misweighted available information.

Interpretation: The design identifies detectable and correctable forecast distortions, not metaphysical irrationality.

5.2 Why conditioning on the survey forecast matters

- Professional forecasters may have private information or judgment not observable in public datasets.
- If the machine ignored the survey forecast, it might falsely label useful private information as bias.
- Conditioning on the current survey forecast allows the machine to control for the private or judgmental component embedded in the human forecast.

Interpretation: The paper's benchmark is designed to be fair to forecasters. It asks whether their own forecast should be reweighted, not whether humans know nothing.

5.3 Threats and limitations

- Percentile types are not individual forecasters, so the results describe positions in the forecast distribution rather than fixed people.
- The machine benchmark depends on the chosen loss function, information set, estimator, and survey timing assumptions.
- The algorithm may miss information that was available but not included in the constructed dataset.
- The machine can still make large ex post errors, especially around unforeseen events such as the Great Recession.
- The results are about macro survey expectations and may not automatically generalize to all economic beliefs.

Interpretation: The paper is careful not to equate the machine with omniscience. Its contribution is to construct a disciplined real-time benchmark.

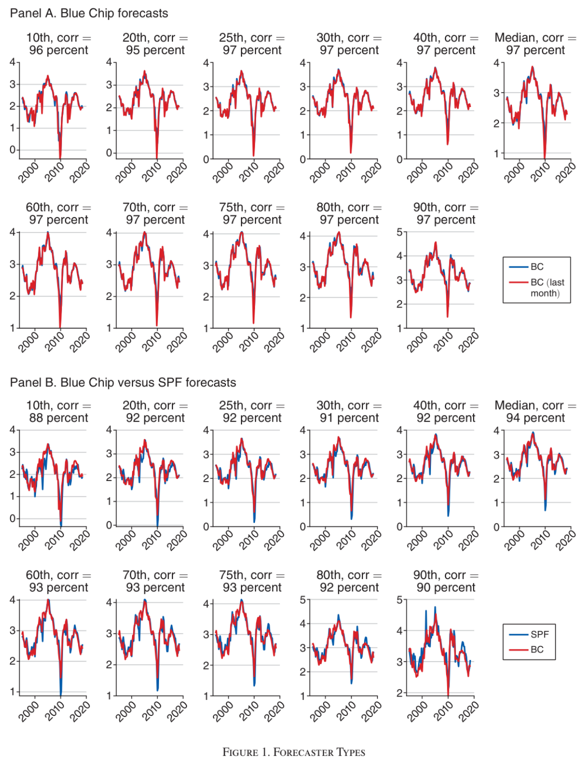
6 Tables and figures

6.1 Figure 1 and Figure 2: Forecaster types and real-time recession thresholds

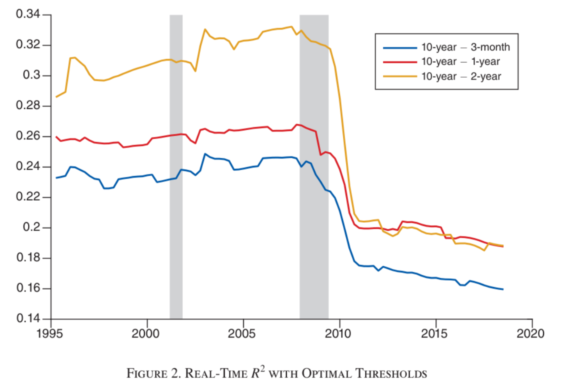
Read:

- Figure 1 compares current professional forecast percentiles with previously available professional forecast distributions.
- Panel A shows that Blue Chip percentiles are highly persistent from one month to the next; correlations are often around 95 to 97 percent.
- Panel B shows that SPF percentiles are also strongly correlated with the most recent Blue Chip percentiles for the same variable and forecast horizon.
- Figure 2 shows real-time R^2 values from threshold regressions using alternative Treasury term spreads.
- The 10-year minus 2-year spread generally has the highest real-time predictive power for GDP growth in the evaluation period.

Economic message: Figure 1 supports the assumption that professional forecasters can approximately know their contemporaneous type in the distribution. Figure 2 explains why the machine uses a yield-curve switching rule for GDP growth: by the mid-1990s, term spreads had well-established real-time predictive content for downturns.



(a) Figure 1: Forecaster types



(b) Figure 2: Real-time R^2 with optimal thresholds

6.2 Table 1: Machine learning versus survey forecasts

Read:

- Table 1 is the paper's core forecast-performance table.
- For inflation, the machine beats every survey and every respondent type: the ratio $MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$ is below one throughout.
- For the median inflation forecast, the ratios are 0.85 for SPF, 0.62 for SOC, and 0.84 for BC.
- For GDP growth, the median ratios are 0.88 for SPF, 0.78 for SOC, and 0.88 for BC.
- Higher percentiles often show even larger improvements, especially for household inflation expectations.
- The hybrid forecast weights are generally well above 0.5, meaning that an optimal combination would place substantial weight on the machine forecast.

Economic message: Table 1 shows that the machine does not merely improve a special case. It improves inflation and GDP forecasts across professional forecasters, households, Blue Chip respondents, and heterogeneous forecast types.

Table 1—Machine Learning versus Survey Forecasts

ML: $y_{j,t+h} = \alpha_{jh}^{(i)} + \beta_{jh\mathbb{F}}^{(i)} \mathbb{F}_t^{(i)}[y_{j,t+h}] + \mathbf{B}_{jh\mathcal{Z}}^{(i)} \mathcal{Z}_{jt} + \epsilon_{jt+h}$

Inflation forecasts

Percentile:	Median	5th	10th	20th	25th	30th	40th
Survey of Professional Forecasters (SPF)							
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.85	0.56	0.74	0.83	0.90	0.88	0.89
OOS R^2	0.15	0.44	0.26	0.17	0.10	0.12	0.11
w^*	0.68	0.82	0.78	0.74	0.63	0.66	0.66
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.82	0.41	0.64	0.77	0.81	0.82	0.84
	60th	70th	75th	80th	90th	95th	
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.74	0.70	0.67	0.59	0.55	0.47	
OOS R^2	0.26	0.30	0.33	0.41	0.45	0.53	
w^*	0.78	0.76	0.74	0.80	0.79	0.83	
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.76	0.66	0.61	0.53	0.39	0.27	
Percentile:	Median	5th	10th	20th	25th	30th	40th
Michigan Survey of Consumers (SOC)							
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.62	0.22	0.27	0.45	0.58	0.69	0.70
OOS R^2	0.38	0.78	0.73	0.55	0.42	0.31	0.30
w^*	1.00	0.96	0.93	0.92	0.92	0.95	1.00
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.62	0.11	0.20	0.42	0.52	0.64	0.76
	60th	70th	75th	80th	90th	95th	
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.40	0.22	0.16	0.13	0.05	0.03	
OOS R^2	0.60	0.78	0.84	0.87	0.95	0.97	
w^*	1.00	1.00	1.00	1.00	1.00	1.00	
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.37	0.21	0.16	0.11	0.04	0.02	
Percentile:	Median	5th	10th	20th	25th	30th	40th
Blue Chip Financial Forecasts (BC)							
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.84	0.58	0.60	0.85	0.85	0.86	0.91
OOS R^2	0.16	0.42	0.40	0.15	0.15	0.14	0.09
w^*	0.65	0.73	0.76	0.62	0.63	0.62	0.58
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.76	0.45	0.55	0.72	0.76	0.78	0.78
	60th	70th	75th	80th	90th	95th	
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.78	0.69	0.65	0.59	0.48	0.38	
OOS R^2	0.22	0.31	0.35	0.41	0.52	0.62	
w^*	0.70	0.79	0.82	0.86	0.94	0.92	
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.73	0.66	0.62	0.57	0.43	0.33	

(*continued*)

(a) Table 1, part 1: Inflation forecasts

Table 1—Machine Learning versus Survey Forecasts (*continued*)

ML: $y_{j,t+h} = \alpha_{jh}^{(i)} + \beta_{jh\mathbb{F}}^{(i)} \mathbb{F}_t^{(i)}[y_{j,t+h}] + \mathbf{B}_{jh\mathcal{Z}}^{(i)} \mathcal{Z}_{jt} + \epsilon_{jt+h}$

GDP forecasts

Percentile:	Median	5th	10th	20th	25th	30th	40th
Survey of Professional Forecasters (SPF)							
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.88	0.70	0.81	0.80	0.84	0.87	0.88
OOS R^2	0.12	0.30	0.19	0.20	0.16	0.13	0.12
w^*	0.83	0.96	1.00	1.00	1.00	0.94	0.87
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.87	0.74	0.83	0.88	0.89	0.89	0.88
	60th	70th	75th	80th	90th	95th	
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.85	0.81	0.79	0.80	0.69	0.64	
OOS R^2	0.15	0.19	0.21	0.20	0.31	0.36	
w^*	0.85	0.88	0.87	0.83	0.85	0.84	
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.84	0.81	0.79	0.77	0.67	0.58	
Percentile:	Median						
Michigan Survey of Consumers (SOC)							
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.78						
OOS R	0.22						
w^*	0.81						
Percentile:	Median	5th	10th	20th	25th	30th	40th
Blue Chip Financial Forecasts (BC)							
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.88	0.77	0.74	0.89	0.83	0.82	0.78
OOS R^2	0.13	0.23	0.26	0.11	0.17	0.18	0.22
w^*	0.64	0.68	0.75	0.60	0.73	0.71	0.76
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.75	0.70	0.76	0.79	0.79	0.78	0.76
	60th	70th	75th	80th	90th	95th	
$MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$	0.78	0.79	0.75	0.71	0.67	0.67	
OOS R^2	0.22	0.21	0.25	0.29	0.33	0.33	
w^*	0.72	0.71	0.72	0.77	0.77	0.74	
$MSE_{\mathbb{R}}/MSE_{\mathbb{F}}$	0.73	0.70	0.69	0.66	0.60	0.55	

Notes: The machine and survey mean squared forecast errors for four-quarter-ahead forecasts, averaged over the evaluation sample are denoted by $\text{MSE}_{\mathbb{E}}$ and $\text{MSE}_{\mathbb{F}}$, respectively. The out-of-sample R^2, OOS R^2, is defined as $1 - \text{MSE}_{\mathbb{E}}/\text{MSE}_{\mathbb{F}}$. The weight on the machine forecast for the hybrid forecast described in the text is denoted by w^*. The MSE from the full information, rational expectations (FIRE) benchmark described in the text is denoted by $\text{MSE}_{\mathbb{R}}$. The vintage of observations on the variable being forecast is the one available four quarters after the period being forecast. The evaluation period for the Survey of Professional Forecasters (SPF) and the Michigan Survey of Consumers (SOC) is 1995:I to 2018:II; and for the Blue Chip (BC) survey is 1997:III to 2018:II (inflation) and 1996:I to 2018:II (GDP).

(b) Table 1, part 2: GDP forecasts

6.3 Table 2 and Figure 3: Recent forecast performance and machine sparsity

Read:

- Table 2 focuses on the last five years of the evaluation sample, 2013:II to 2018:II.
- The machine improvement is larger at the end of the sample than over the full sample.
- For median inflation, the last-five-year ratios are 0.63 for SPF, 0.37 for SOC, and 0.67 for BC.
- For median GDP growth, the last-five-year ratios are 0.82 for SPF, 0.71 for SOC, and 0.67 for BC.
- Figure 3 shows that the elastic-net algorithm often chooses sparse models, but not the same sparse model every time.
- The machine sometimes uses strong shrinkage, strong sparsity, or both, depending on the period and forecast target.

Economic message: The biggest errors occurred when information-processing technology was already widely available, which weakens the interpretation that distortions are driven only by fixed human capacity limits. Figure 3 shows why a dynamic algorithm matters: the relevant information set changes over time.

TABLE 2—MACHINE LEARNING VERSUS SURVEY FORECASTS

$$\text{ML: } y_{j,t+h} = \alpha_{jh}^{(i)} + \beta_{jh\mathbb{F}}^{(i)} \mathbb{F}_t^{(i)}\left[y_{j,t+h}\right] + \mathbf{B}_{jh\mathcal{Z}}^{(i)} \mathcal{Z}_{jt} + \epsilon_{jt+h}$$

Median inflation forecasts, $MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$

SPF		SOC		BC	
1995:I–2018:II	2013:II–2018:II	1995:I–2018:II	2013:II–2018:II	1997:III–2018:II	2013:II–2018:II
0.85	0.63	0.62	0.37	0.84	0.67

Median GDP forecasts, $\mathrm{MSE}_{\mathbb{E}}/\mathrm{MSE}_{\mathbb{F}}$

SPF		SOC		BC	
1995:I–2018:II	2013:II–2018:II	1995:I–2018:II	2013:II–2018:II	1996:I–2018:II	2013:II–2018:II
0.88	0.82	0.78	0.71	0.87	0.67

Notes: Machine versus survey mean squared forecast errors. The machine and survey mean squared forecast errors, for four-quarter-ahead forecasts, averaged over the evaluation sample are denoted by $\mathrm{MSE}_{\mathbb{E}}$ and $\mathrm{MSE}_{\mathbb{F}}$, respectively. The vintage of observations on the variable being forecast is the one available four quarters after the period being forecast.

(a) Table 2: Machine performance in recent years

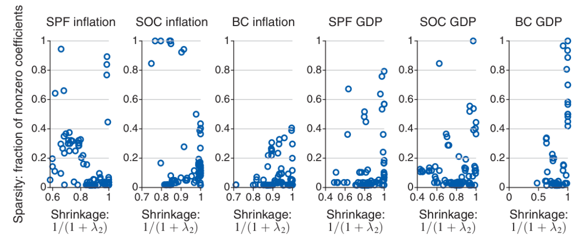


FIGURE 3. DEGREE OF SPARSITY AND SHRINKAGE

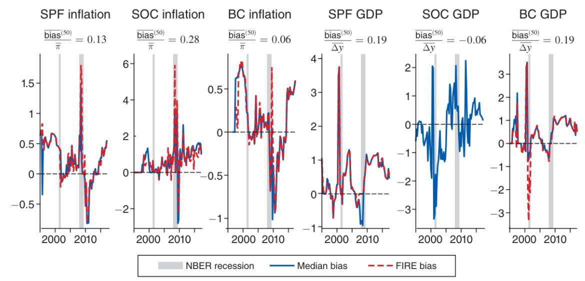
(b) Figure 3: Degree of sparsity and shrinkage

6.4 Figure 4 and Figure 5: Bias dynamics and heterogeneity

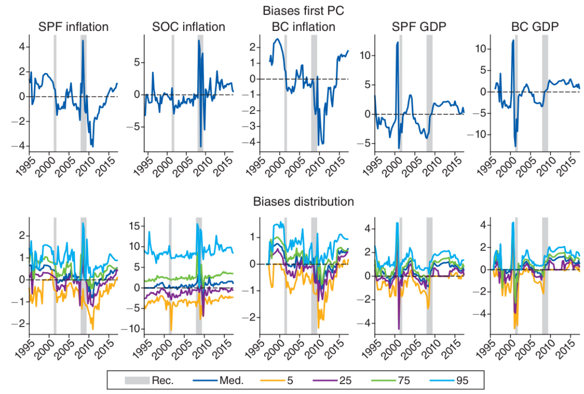
Read:

- Figure 4 plots the median survey bias relative to the machine benchmark for inflation and GDP growth.
- Inflation expectations are generally biased upward for SPF and SOC, while BC inflation bias is closer to zero on average.
- GDP growth expectations of professional forecasters are often optimistic after the Great Recession.
- Figure 5 separates common and heterogeneous components of belief distortions.
- The common component moves substantially over time, but there is also large cross-sectional disagreement across respondent percentiles.
- Household inflation beliefs display especially large dispersion.

Economic message: Belief distortions are not constant biases. They move in waves and differ across respondents. The macroeconomy contains both common movements in beliefs and substantial disagreement or "noise."



(a) Figure 4: Biases in the median survey forecast



(b) Figure 5: Common and heterogeneous distortions

6.5 Figure 6 and Figure 7: Forecasted versus actual outcomes and bias decomposition

Read:

- Figure 6 compares median survey forecasts, machine forecasts, and realized outcomes.
- In the last part of the sample, professional forecasters and households repeatedly overpredict inflation and growth.
- The machine also misses the Great Recession in real time, showing that large ex post errors can arise from unforeseen events rather than systematic bias alone.
- Figure 7 decomposes bias into three parts: the weight on the survey forecast, the rolling mean, and public information variables.
- The red bars in the survey-forecast row show that respondents frequently overweight their own forecast.
- The public information variables that are underweighted change over time: credit spreads, Treasury information, lagged survey forecasts, trend inflation, and employment factors matter in different periods.

Economic message: Figure 7 contains one of the paper's central behavioral findings: forecasters' judgment contains useful information, but it is overweighted. Human forecasts can be improved by systematically downweighting judgment and upweighting public data.

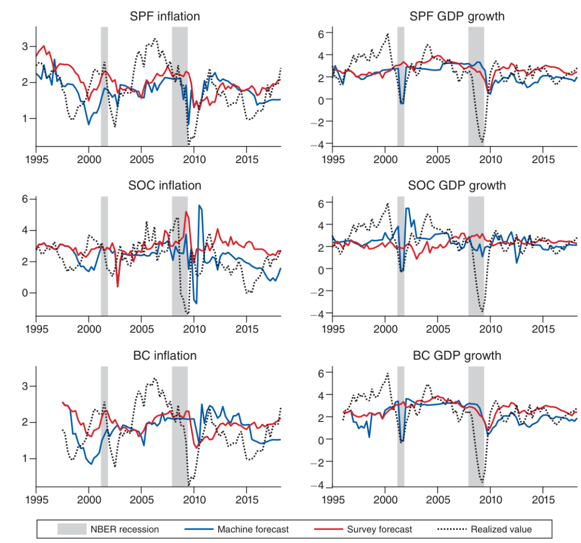


Figure 6. Forecasted versus Actual Inflation, GDP Growth

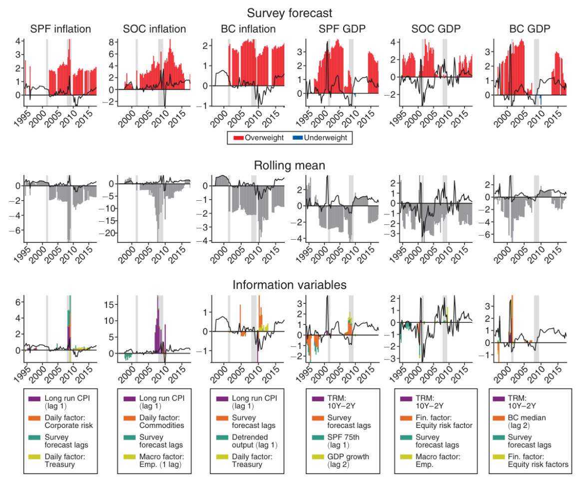


Figure 7. Bias Decomposition: Median Forecast

(a) Figure 6: Forecasted versus actual inflation and GDP growth

(b) Figure 7: Bias decomposition for the median forecast

6.6 Table 3 and Figure 8: Revisiting forecast revisions

Read:

- Table 3 revisits the Coibion-Gorodnichenko forecast-error-on-revision regression.
- In-sample, lagged SPF forecast revisions predict forecast errors, reproducing the familiar evidence associated with information rigidity.
- Out of sample, the same specifications perform worse than the survey forecast itself; the MSE ratios are greater than one across rolling and recursive windows.
- Figure 8 shows that the machine usually shrinks the coefficient on forecast revisions to zero when it can also use other information.
- The in-sample coefficient is around 1.2 in long samples, but the machine rarely finds forecast revisions useful in real-time prediction.

Economic message: The figure and table show why real-time out-of-sample evaluation matters. A pattern that looks strong in-sample can be unhelpful for actual forward-looking forecasting.

TABLE 3—CG REGRESSIONS OF FORECAST ERRORS ON FORECAST REVISIONS

Panel A. In-sample regressions (CG sample)

Regression: $\pi_{t+3} - \mathbb{F}_t^{(\mu)}[\pi_{t+3}] = \alpha^{(\mu)} + \beta^{(\mu)}\left(\mathbb{F}_t^{(\mu)}[\pi_{t+3}] - \mathbb{F}_{t-1}^{(\mu)}[\pi_{t+3}]\right) + \delta\pi_{t-1,t-2} + \epsilon_t$

Constant	0.001	−0.077
t-stat	(0.005)	(−0.442)
$\mathbb{F}_t[\pi_{t+3,t}] - \mathbb{F}_{t-1}[\pi_{t+3,t}]$	1.194	1.141
t-stat	(2.496)	(2.560)
$\pi_{t-1,t-2}$		0.021
t-stat		(0.435)
$\bar{R}^2$	0.195	0.197

Panel B. Out-of-sample regressions

Regression: $\pi_{t+3} - \mathbb{F}_t^{(\mu)}[\pi_{t+3}] = \alpha^{(\mu)} + \beta^{(\mu)}\left(\mathbb{F}_t^{(\mu)}[\pi_{t+3}] - \mathbb{F}_{t-1}^{(\mu)}[\pi_{t+3}]\right) + \epsilon_{t+3}$

Method	Forecast sample	$\mathrm{MSE}_{CG}/\mathrm{MSE}_{\mathbb{F}}$
Rolling 5 years	1975:IV–2018:II	1.38
Rolling 10 years	1980:IV–2018:II	1.29
Rolling 20 years	1990:IV–2018:II	1.31
Recursive 5 years	1975:IV–2018:II	1.69
Recursive 10 years	1980:IV–2018:II	1.60
Recursive 20 years	1990:IV–2018:II	1.33

Notes: Panel A reports the in-sample results over the sample used in Coibion and Gorodnichenko (2015) (CG), 1969:I to 2014:IV. Newey-West corrected *t*-statistics with lags = 4 are reported in parentheses. Panel B reports the ratio of out-of sample MSE of the CG model forecast to that for the survey forecast computed using different rolling or recursive estimation windows. The MSE for the CG model averages the (square of the) forecast errors $\pi_{t+3} - \widehat{\pi}_{t+3}^{(\mu)}$, where $\widehat{\pi}_{t+3}^{(\mu)} = \widehat{\alpha}_t^{(\mu)} + \left(1 + \widehat{\beta}_t^{(\mu)}\right)\mathbb{F}_t^{(\mu)}[\pi_{t+3}] - \widehat{\beta}_t^{(\mu)}\mathbb{F}_{t-1}^{(\mu)}[\pi_{t+3}]$. In both panels, the regression estimation uses the latest vintage of inflation in real time and, following CG, computes forecast errors with real-time data available four quarters after the period being forecast. Annual inflation is defined as $\pi_{t+3,t} = \frac{P_t}{P_{t-1}} \times \frac{P_{t+1}}{P_t} \times \frac{P_{t+2}}{P_{t+1}} \times \frac{P_{t+3}}{P_{t+2}}$, and $\mathbb{F}_t[\pi_{t+3,t}]$ is the mean forecast of annual inflation as of time t from the SPF. The sample of panel B spans the period 1969:I–2018:II.

(a) Table 3: CG regressions of forecast errors on forecast revisions

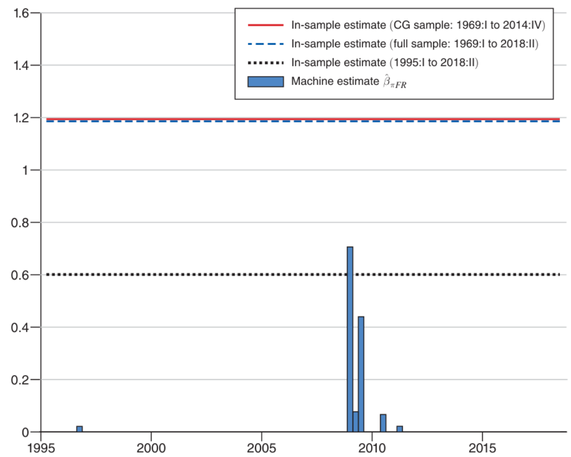


FIGURE 8. COEFFICIENT ON FORECAST REVISIONS

(b) Figure 8: Coefficient on forecast revisions

6.7 Table 4 and Figure 9: Low-dimensional versus high-dimensional models and cyclical shocks

Read:

- Table 4 compares autoregressive lag coefficients in high-dimensional and low-dimensional GDP-growth forecasting specifications.
- In the high-dimensional model, the first two autoregressive coefficients are essentially zero.
- In the low-dimensional model, the autoregressive coefficients receive nonzero weights because fewer other information variables are available.
- Figure 9 studies impulse responses to cyclical shocks identified in prior work.
- The machine improves over the SPF median in this context too, with MSE ratios of 0.60 for inflation and 0.73 for GDP growth.
- Survey beliefs tend to underreact initially and then overreact later, but the magnitude is smaller when compared to the machine benchmark rather than to realized outcomes.

Economic message: Table 4 shows that conclusions about useful predictors can change when the information environment is high-dimensional. Figure 9 shows that belief distortions move with business-cycle shocks, but also that realized outcomes contain surprises that neither humans nor machines could fully forecast in real time.

TABLE 4—AVERAGE COEFFICIENTS ON THE FIRST TWO AR LAGS

	High dimensional	Low dimensional
β_1	0.000	0.022
β_2	-0.002	-0.013

Notes: This table reports average autoregressive coefficients from one-year-ahead rolling regressions of real GDP growth on predictors. The average coefficient on the first AR lag is β_1; the average coefficient on the second is β_2. The high dimension estimation entertains very large numbers of potential predictors, in addition to the autoregressive lags, while the low dimension setting uses only two additional predictors. The sample spans 1995:I–2018:II.

(a) Table 4: Average coefficients on the first two AR lags

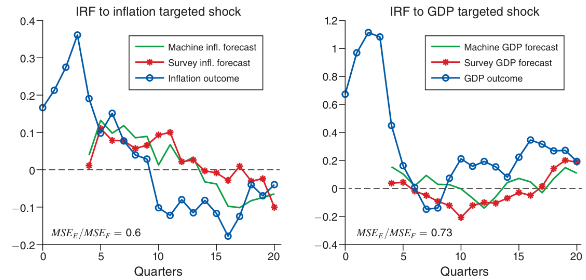


FIGURE 9. DYNAMIC RESPONSES TO CYCLICAL SHOCKS

(b) Figure 9: Dynamic responses to cyclical shocks

7 Short summary

- The paper estimates systematic expectational errors in survey forecasts using a dynamic machine-learning benchmark.
- The benchmark uses only information that was available in real time before the survey deadline.
- The machine is allowed to condition on the current survey forecast, which controls for private information and judgment embedded in the human forecast.
- The machine beats survey forecasts of inflation and GDP growth across SPF, SOC, and BC surveys.
- The machine improvement is especially large from 2013:II to 2018:II.
- Median inflation forecasts are generally biased upward; professional forecasts of GDP growth are often optimistic after the Great Recession.
- The largest and most robust source of bias is overreliance on the private or judgmental component of the survey forecast.
- The public information that forecasters underweight changes over time, supporting the need for a dynamic algorithm.
- Prior in-sample evidence on forecast revisions is much weaker when evaluated out of sample.
- The machine also makes large errors around unforeseen events, which shows that ex post mistakes are not the same as ex ante belief distortions.

8 Conclusion

8.1 Core conclusion

Bianchi, Ludvigson, and Ma show that survey expectations contain economically meaningful and time-varying distortions. A dynamic machine-learning algorithm, trained only on real-time information, consistently improves on human forecasts of inflation and GDP growth. This is true even for professional forecasters.

8.2 Economic intuition

The main intuition is that macroeconomic forecasting is a high-dimensional problem. Humans, including experts, combine public information with private judgment. That judgment is often useful, but the paper shows that respondents generally put too much relative weight on it. A disciplined machine can reduce these errors by reweighting judgment against large-scale public information.

8.3 Incremental contribution revisited

The paper contributes by replacing an abstract rational-expectations benchmark with a real-time machine-efficient benchmark. It avoids hindsight, uses large information sets, and treats forecasting as a genuine out-of-sample decision problem. This changes the interpretation of several earlier behavioral-macro findings and produces more conservative measures of belief distortion.

8.4 Implications for macroeconomics

The results provide empirical support for macro theories in which expectational errors matter. However, they also suggest that simple stories of fixed underreaction or fixed overreaction are incomplete. Distortions vary over time, differ across respondent types, and depend on which information is underweighted in a given period.

8.5 Implications for forecasting practice

Forecasting teams can use machine learning as an audit tool. The algorithm need not replace experts, but it can discipline expert judgment by checking whether the forecast overweights private impressions relative to data. The paper's approach is therefore practically relevant for central banks, financial institutions, and macroeconomic forecasters.

8.6 Limitations

The machine benchmark is not a proof of true rational expectations. It depends on the included information set, the elastic-net method, the loss function, and the percentile-type construction. The analysis also studies forecasts rather than the full decision processes of agents. Finally, the Great Recession illustrates that even a data-rich machine can miss major events in real time.

8.7 Takeaway

The enduring takeaway is that belief distortions are measurable, economically meaningful, and dynamically related to macroeconomic fluctuations. Human forecasts contain valuable judgment, but that judgment is often overweighted. A real-time machine benchmark can help identify where human expectations deviate from a more disciplined use of available information.